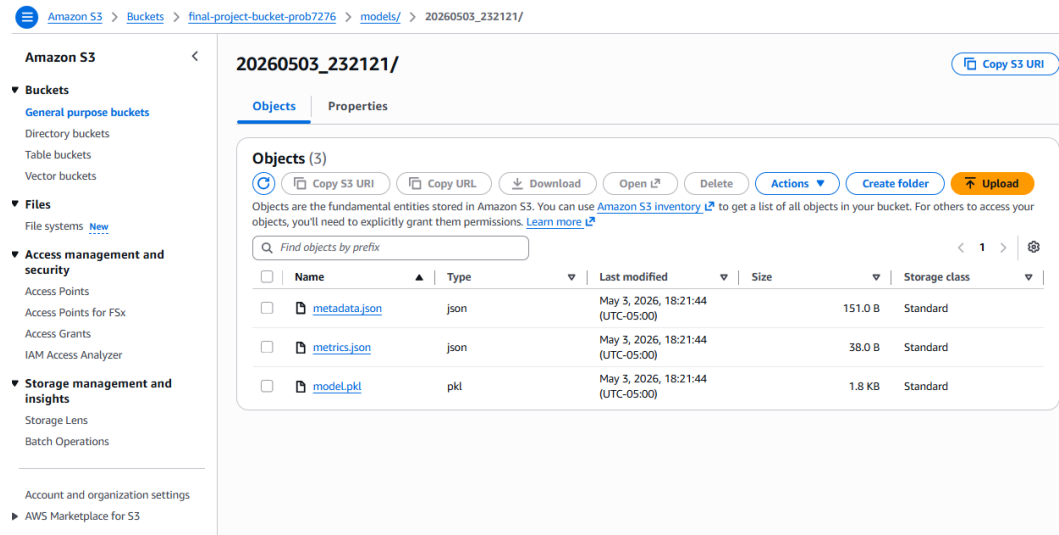
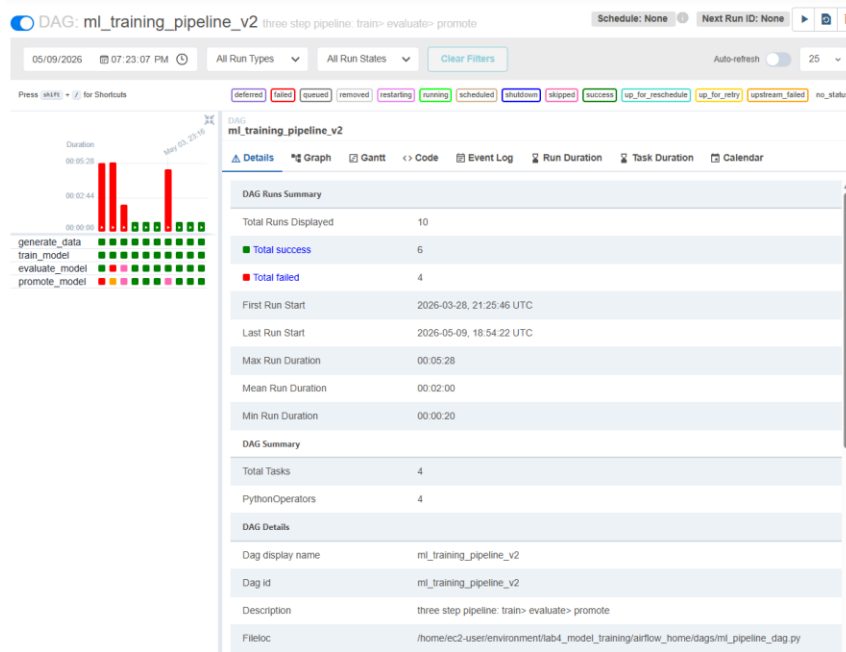


Jacob Probst
MLOPs Final Project/HW 5
SIES 765
https://github.com/prob7276/MLOPS_Final_Project

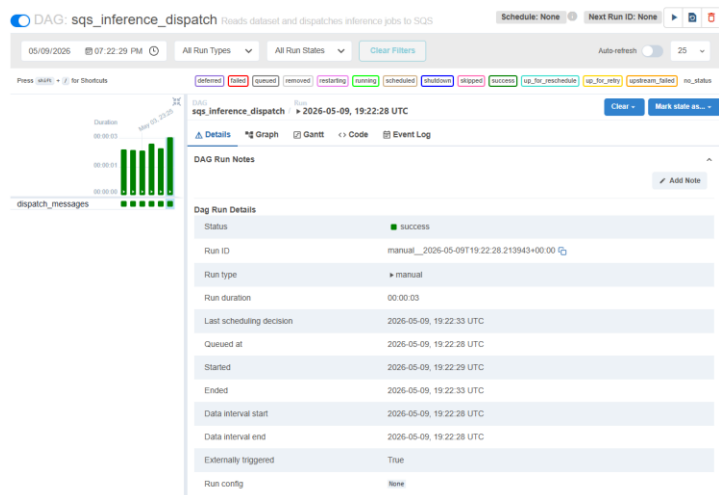
1. Airflow Training DAG

model.pkl stored in S3



2. Airflow Queue Population

Each message must include: { "record_id": "sample_001", "features": [...] }



3. Kubernetes Consumer

MLOPS-cluster2

Connect

Delete cluster

Monitor cluster

▼ Cluster info info

Status
Active

Kubernetes version
1.35

Support period
Standard support until
March 26, 2027

Provider
EKS

Cluster health
0

Upgrade insights
5

Node health issues
0

Capability issues
0

< Overview Resources Compute Networking Add-ons 1 Capabilities Access Observability UI >

Nodes (2) info

Q Filter Nodes by property or value

< 1 >

Node name	Instance type	Compute	Managed by	Created	Status	CPU usage	Memory usage
ip-172-31-0-102.ec2.internal	t3.medium	Node group	mlops-workers	an hour ago	Ready	12%	15%
ip-172-31-26-237.ec2.internal	t3.medium	Node group	mlops-workers	an hour ago	Ready	32%	30%

Node groups (1) info

Edit Delete Add

Node groups implement basic compute scaling through EC2 Auto Scaling groups.

Q Filter node groups by property or value

< 1 >

Group name	Desired size	AMI release version	Launch template	Status
mlops-workers	2	1.35.4-20260505	-	Active

Fargate profiles (0) info

Edit Delete Add

Q Filter Fargate profiles by property or value

< 1 >

Profile name	Namespaces	Status
No Fargate profiles This cluster does not have any Fargate profiles.		

Add Fargate profile

3

4. Output Requirements

The screenshot shows the Amazon S3 console interface. On the left, the navigation pane includes sections for Buckets, Files, Access management and security, Storage management and insights, and Account and organization settings. The main content area displays the 'predictions/' bucket with a list of objects. A preview window for 'sample_000.json' is open, showing the following JSON content:

```
{"record_id": "sample_000", "prediction": 0, "timestamp": "2026-05-09T19:24:09.599179+00:00"}
```

Below the preview window, a table lists the objects in the bucket:

Name	Type	Last modified	Size	Storage class
sample_000.json	json	May 9, 2026, 14:24:11 (UTC-05:00)	93.0 B	Standard
sample_011.json	json	May 9, 2026, 14:24:11 (UTC-05:00)	93.0 B	Standard
sample_012.json	json	May 9, 2026, 14:24:11 (UTC-05:00)	93.0 B	Standard
sample_013.json	json	May 9, 2026, 14:24:11 (UTC-05:00)	93.0 B	Standard
sample_014.json	json	May 9, 2026, 14:24:11 (UTC-05:00)	93.0 B	Standard

5. Kubernetes Deployment

```
voclabs:~/environment/lab4_model_training/consumer (main) $ kubectl get deployment inference-worker-deployment
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
inference-worker-deployment        0/5      5              0            69m
voclabs:~/environment/lab4_model_training/consumer (main) $
```

bash: dockerfile: command not found

```
voclabs:~/environment/lab4_model_training/consumer (main) $ kubectl get pods -w
NAME                                READY    STATUS              RESTARTS    AGE
inference-worker-deployment-5dd7f5796b-5rhq7  0/1      CrashLoopBackOff    10 (2m28s ago)  30m
inference-worker-deployment-5dd7f5796b-6hlxb  0/1      CrashLoopBackOff    10 (2m42s ago)  30m
inference-worker-deployment-5dd7f5796b-g6x72  0/1      CrashLoopBackOff    10 (3m21s ago)  30m
inference-worker-deployment-5dd7f5796b-14cfq  0/1      CrashLoopBackOff    10 (2m46s ago)  30m
inference-worker-deployment-5dd7f5796b-mwb8g  0/1      CrashLoopBackOff    15 (4m28s ago)  58m
inference-worker-deployment-5dd7f5796b-mwb8g  1/1      Running             16 (5m14s ago)  59m
inference-worker-deployment-5dd7f5796b-mwb8g  0/1      Completed          16 (5m22s ago)  59m
inference-worker-deployment-5dd7f5796b-g6x72  1/1      Running             11 (5m2s ago)   32m
inference-worker-deployment-5dd7f5796b-g6x72  0/1      Completed          11 (5m10s ago)  32m
inference-worker-deployment-5dd7f5796b-mwb8g  0/1      CrashLoopBackOff    16 (87s ago)    61m
inference-worker-deployment-5dd7f5796b-14cfq  1/1      Running             11 (5m12s ago)  33m
inference-worker-deployment-5dd7f5796b-14cfq  0/1      Completed          11 (5m19s ago)  33m
inference-worker-deployment-5dd7f5796b-5rhq7  1/1      Running             11 (5m6s ago)   33m
inference-worker-deployment-5dd7f5796b-5rhq7  0/1      Completed          11 (5m13s ago)  33m
inference-worker-deployment-5dd7f5796b-6hlxb  1/1      Running             11 (5m27s ago)  33m
inference-worker-deployment-5dd7f5796b-6hlxb  0/1      Completed          11 (5m34s ago)  33m
inference-worker-deployment-5dd7f5796b-g6x72  0/1      CrashLoopBackOff    11 (72s ago)    33m
inference-worker-deployment-5dd7f5796b-14cfq  0/1      CrashLoopBackOff    11 (74s ago)    34m
inference-worker-deployment-5dd7f5796b-5rhq7  0/1      CrashLoopBackOff    11 (87s ago)    34m
inference-worker-deployment-5dd7f5796b-6hlxb  0/1      CrashLoopBackOff    11 (82s ago)    34m
inference-worker-deployment-5dd7f5796b-mwb8g  1/1      Running             17 (5m17s ago)  64m
```

```
voclabs:~/environment/lab4_model_training/consumer (main) $ kubectl logs -l app=inference-worker -f --prefix
[pod/inference-worker-deployment-5dd7f5796b-mwb8g/inference-worker] message DELETED!!!
[pod/inference-worker-deployment-5dd7f5796b-mwb8g/inference-worker] Processing sample_039...
[pod/inference-worker-deployment-5dd7f5796b-mwb8g/inference-worker] saved to s3://final-project-bucket-prob7276/predictions/sample_039.json
[pod/inference-worker-deployment-5dd7f5796b-g6x72/inference-worker] downloading...
[pod/inference-worker-deployment-5dd7f5796b-g6x72/inference-worker] success
[pod/inference-worker-deployment-5dd7f5796b-g6x72/inference-worker] SQS processing meassages
[pod/inference-worker-deployment-5dd7f5796b-g6x72/inference-worker] Queue is empty...
[pod/inference-worker-deployment-5dd7f5796b-mwb8g/inference-worker] message DELETED!!!
[pod/inference-worker-deployment-5dd7f5796b-mwb8g/inference-worker] Processing sample_049...
[pod/inference-worker-deployment-5dd7f5796b-mwb8g/inference-worker] saved to s3://final-project-bucket-prob7276/predictions/sample_049.json
[pod/inference-worker-deployment-5dd7f5796b-mwb8g/inference-worker] message DELETED!!!
[pod/inference-worker-deployment-5dd7f5796b-mwb8g/inference-worker] Queue is empty...
[pod/inference-worker-deployment-5dd7f5796b-5rhq7/inference-worker] downloading...
[pod/inference-worker-deployment-5dd7f5796b-5rhq7/inference-worker] success
[pod/inference-worker-deployment-5dd7f5796b-5rhq7/inference-worker] SQS processing meassages
[pod/inference-worker-deployment-5dd7f5796b-5rhq7/inference-worker] Queue is empty...
[pod/inference-worker-deployment-5dd7f5796b-14cfq/inference-worker] downloading...
[pod/inference-worker-deployment-5dd7f5796b-14cfq/inference-worker] success
[pod/inference-worker-deployment-5dd7f5796b-14cfq/inference-worker] SQS processing meassages
[pod/inference-worker-deployment-5dd7f5796b-14cfq/inference-worker] Queue is empty...
[pod/inference-worker-deployment-5dd7f5796b-6hlxb/inference-worker] downloading...
[pod/inference-worker-deployment-5dd7f5796b-6hlxb/inference-worker] success
[pod/inference-worker-deployment-5dd7f5796b-6hlxb/inference-worker] SQS processing meassages
[pod/inference-worker-deployment-5dd7f5796b-6hlxb/inference-worker] Queue is empty...
voclabs:~/environment/lab4_model_training/consumer (main) $
```

```
voclabs:~/environment/lab4_model_training/consumer (main) $ aws s3 ls s3://final-project-bucket-prob7276/predictions/ --human-readable --summarize
2026-05-09 19:50:54 93 Bytes sample_000.json
2026-05-09 19:50:55 93 Bytes sample_001.json
2026-05-09 19:50:55 93 Bytes sample_002.json
2026-05-09 19:50:55 93 Bytes sample_003.json
2026-05-09 19:50:55 93 Bytes sample_004.json
2026-05-09 19:50:55 93 Bytes sample_005.json
2026-05-09 19:50:54 93 Bytes sample_006.json
2026-05-09 19:50:56 93 Bytes sample_007.json
2026-05-09 19:50:55 93 Bytes sample_008.json
2026-05-09 19:50:55 93 Bytes sample_009.json
2026-05-09 19:50:55 93 Bytes sample_010.json
2026-05-09 19:50:54 93 Bytes sample_011.json
2026-05-09 19:50:54 93 Bytes sample_012.json
2026-05-09 19:50:55 93 Bytes sample_013.json
2026-05-09 19:50:56 93 Bytes sample_014.json
2026-05-09 19:50:55 93 Bytes sample_015.json
2026-05-09 19:50:56 93 Bytes sample_016.json
2026-05-09 19:50:56 93 Bytes sample_017.json
2026-05-09 19:50:54 93 Bytes sample_018.json
2026-05-09 19:50:56 93 Bytes sample_019.json
```